5 **(a)** A straight rigid conducting wire of mass 0.14 kg and length 0.40 m is placed horizontally on a frictionless plastic plane. The plane is at an angle of 20° to the horizontal and is placed in a region of upwards uniform vertical magnetic field as shown in Fig. 5.1 and Fig. 5.2.

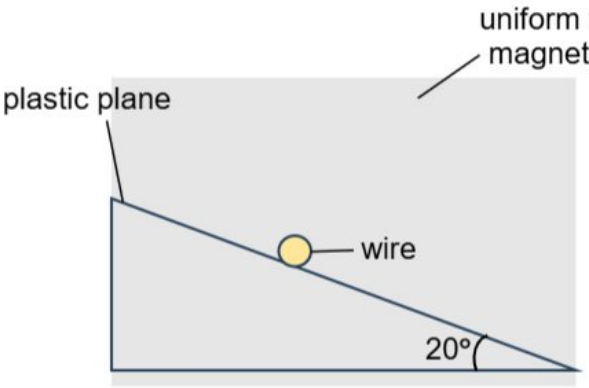


Fig. 5.1 (side view)

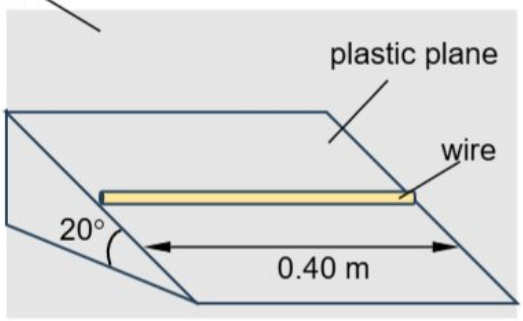


Fig. 5.2

To prevent the wire from sliding down the plane, a steady current I is passed through the wire so that it remains stationary on the plastic plane.

(i) On Fig. 5.1, draw an arrow to show the direction of the magnetic force acting on the wire. [1]

(ii) On Fig. 5.2, draw an arrow to show the direction of the current I in the wire. [1]

(iii) With reference to forces acting on the wire, explain why the wire remains stationary.

.....

.....

.....

.....

.....[2]

(iv) Suggest and explain whether the wire will eventually slide up, remain stationary or slide down the plane if an alternating current of root-mean-square value of I is passed through the wire instead.

.....

.....

.....

.....

.....[2]

(b) The current in **(a)** is disconnected, and the magnetic field is removed. The wire is now given a uniform net positive charge of $+2.0 \times 10^{-6}$ C.

(i) To keep the wire stationary in the same position on the frictionless plane, a uniform electric field of strength E is applied in the horizontal direction.

Determine the magnitude of E.

E = ... N C^{-1} [3]

[Total: 9]